6 TESTS OF HYPOTHESES

INTRODUCTION

A hypothesis is an assumption about the population parameter to be tested based on sample information.

Procedure of Hypothesis Testing

The general procedure followed in testing hypothesis comprises the following steps:

(1) Set up a hypothesis

The first step in hypothesis testing is to establish the hypothesis to be tested. Since statistical hypothesis are usually assumptions about the value of some unknown parameter, the hypothesis specifies a numerical value or range of values for the parameter. The conventional approach to hypothesis testing is not to construct single hypothesis about the population parameter, but rather to set up two different hypothesis. These hypothesis are normally referred to as

(i) null hypothesis denoted by H_0 - A hypothesis which states that there is no difference between the assumed and actual value of the parameter is the null hypothesis. A hypothesis that is different from the null hypothesis is the alternative hypothesis.

(ii) Alternative hypothesis denoted by H_1 . the alternative hypothesis specifies those values that the researcher believes to hold true.

If the sample information leads us to reject H_0 , then we will accept the alternative hypothesis H_1.

(2) Set up a suitable significance level

Select a suitable level of significance usually denoted by "α". The confidence with which an experimenter rejects or retains null hypothesis depends on the significance level adopted. When the null hypothesis is rejected at $\propto = 0.05$, the test result is said to be "significant". When the null hypothesis is rejected at $\propto = 0.01$, the test result is said to be "highly significant".

(3) Determination of a suitable test statistic

The third step is to determine a suitable test statistic and its distribution. Many of the test statistics that that we shall encounter will be of the following form:

$$\text{Test statistic} = \frac{Sample\ \ Statistic\ \ - Hypot\ hesised\ \ population\ \ \ parameter}{standard\ \ \ error\ \ of\ \ the\ sample\ \ ststistic}$$

(4) Determine the critical region

It is important to specify, before the sample is taken, which value of the test statistic will lead to a rejection of H_0.The former is called the critical region. The value of α , the level of significance, indicates the importance that one attaches to the consequences associated with incorrectly rejecting H_0 . It can be shown that the level of significance is α , the optimal critical region for a two sided test consists of that $\propto/2$ per cent in the right- hand tail of the distribution plus that $\propto/2$

per cent in the left hand tail. Thus, establishing a critical region is similar to determining a 100 (1- $\propto$)% confidence interval. In general, one uses a level of significance of $\propto = 0.05$, indicating that one is willing to accept a 5 per cent chance of being wrong to reject H_0.

(5) Doing Computations

The fifth step in testing hypothesis is the performance of various computations from a random sample of size n, necessary for the test statistic obtained in step (3). Then, we need to see whether sample result falls in the critical region or in the acceptance regions.

(6) Making decisions

Finally, we may draw statistical conclusions and the management may take decisions. A statistical decision or conclusion comprises either accepting the null hypothesis or rejecting it. The decision will depend on whether the computed value of the test criterion falls in the region of rejection or the region of acceptance. If the hypothesis is being tested at 5 per cent level of significance and the observed set of results has a probability less than 5 per cent, we reject the null hypothesis and the difference between the sample statistic and the hypothetical population parameter is considered to be significant. On the other hand, if the testing statistic fall in the region of non-rejection, the null hypothesis is accepted and the difference between the sample statistic and the hypothetical population parameter is not regarded as significant, i.e. it can be explained by chance variations.

Type I and Type II Errors

When a statistical hypothesis is tested, there are four possible results:

1) The hypothesis is true but our test rejects it.
2) The hypothesis is false but our test accepts it.

3) The hypothesis is true but our test accepts it.

4) The hypothesis is false and our test rejects it.

Obviously, the first two possibilities lead to errors. If we reject a hypothesis when it should be accepted (possibility No. 1), we say that a *Type 1 error* has been made. On the other hand, if we accept a hypothesis when it should be rejected (possibility No.2), we say that a *Type 11error* has been made. In either case a wrong decision or error in judgement has occurred.

TWO KINDS OF ERRORS IN HYPOTHESIS TESTING

Decision	Condition: H_0: True	Condition: H_o: False
Accept H_0	Correct decision	Type 11 Error
Reject H_0	Type 1 Error	Correct Decision

The probability of committing a type 1 error is designated as " $\propto$" and is called the level of significance.

Therefore,

$$\propto = P_r \text{ [Type 1 error]}$$

$$= p_r \text{ [Rejecting } H_0/H_0 \text{ is true]}$$

Must be the complement of

$$(1-\alpha) = p_r \text{ [Accepting } H_0/H_0 \text{ is true].}$$

This probability $(1-\alpha)$ corresponds to the concept of $100(1-\alpha)\%$ confidence interval. Our efforts would obviously be to have a small probability of making a type 1 error. Hence the objective is to construct the test to minimize $\propto$.

Similarly, the probability of committing a type 11 error is designated by β. Thus

$$\beta = P_r \text{ [Type 11 error]}$$

$$= p_r \text{ [Accepting } H_0 / H_0 \text{ is false]}$$

and $(1-\beta) = p_r$ [Rejecting H_0 / H_0 is false]

This probability $(1-\beta)$ is known as the power of a statistical test.

The following table gives the probabilities associated with each of the four cells shown in the previous table:

The decision is:	The null hypothesis is	
	True	False
Accept H_0	$(1-\alpha)$ Confidence level	β
Reject H_0	α	$(1-\beta)$ Power of the test
Sum	1.00	1.00

Note that the probability of each decision outcome is a conditional probability and the elements in the same column sum to 1.0, since the events with which they are associated are complement. However, α and β are not independent of each other, nor are they independent of the sample size n. When n is fixed, if α is lowered then β normally rises and vice versa. If n is increased, it is possible for both α and β to decrease. Since increasing the sample size involves money and time, therefore, one should decide how much additional money and time , he is willing to

spare on increasing the sample size in order to reduce the size of α and β .

In order for any tests of hypothesis or rules of decisions to be good, they must be designed so as to minimize errors of decision. However, this is not as simple matter, since for a given sample size, an attempt to decrease one type of error is accompanied in general by an increase in other type of error. The probability of making type 1 error is fixed in advance by the choice of level of significance employed in the test. We can make the type 1 error as small as we please, by lowering the level of significance. But, by doing so, we increase the chance of accepting a false hypothesis, *i.e.,* of making a type II error. It follows that it is impossible to minimize both errors simultaneously. In the long run, errors of type 1 are perhaps more likely to prove serious in research programmes in social sciences than are errors of type II. In practice, one type of error may be more serious than the other and so a compromise should be reached in favour of limitations of the more serious error. The only way to reduce both types of error is to increase the sample size which may not be possible.

One –Tailed and Two Tailed Tests

Basically, there are three kinds of problems of tests of hypothesis. They include:

i. Two -tailed tests ii. Right- tailed test and (iii) Left- tailed test.

Two- tailed test is that where the hypothesis about the population mean is rejected for value of falling into either tail of the sampling distribution. When the hypothesis about population mean is rejected only for value of falling into one of the tails of the sampling distribution, then it is known

as one-tailed test. If, it is right tail then it is called right -tailed test or one–sided alternative to the right and if it is on the left tail, then it is one-sided alternative to the left called left-tailed test. For example, $H_0 : \mu = 100$ tested against $H_1 : \mu > 100 \; or \; < 100$ is one tailed test since H_1 specifies that μ lies on particular side of 100. The same null hypothesis tested against $H_1 : \mu \neq 100$ is a two tailed test since μ lies on a particular side of 100. The following diagram would make it clearer:

The following table gives critical values of z for both one-tailed and two tailed tests at various levels of significance. Critical values of z for other levels of significance are found by use of the table of normal curve areas.

Level of significance	0.10	0.05	0.01	0.005	0.0002
Critical value of z for one tailed tests	-1.28 or 1.28	-1.645 or 1.645	-2.33 or 2.33	-2.58 or 2.58	-2.88 or 2.88
Critical value of z for two tailed tests	-1.645 and 1.645	-1.96 and 1.96	-2.58 and 2.58	-2.81 and 2.81	-3.08 and 3.08

Tests of Hypothesis Concerning Large Samples

Though it is difficult to draw a clear cut line of demarcation between large and small samples, it is generally agreed that if the size of sample exceeds 30, it should be regarded as large sample. The tests of significance used for large samples are based on the following assumptions:

1) The sampling distribution of a sample statistic is approximately normal.
2) Values given by the samples are sufficiently close to the population value and can be used in its place for the standard error of the estimate.

Thus, we have seen that the normal distribution plays a vital role in tests of hypothesis based on large samples (central limit theorem).

Suppose $\hat{\theta}$ is an unbiased estimate of θ, the population parameter. On the basis of $\hat{\theta}$, taken from sample observations, we want to test the hypothesis whether the sample is drawn from a population whose parameter value is θ, i.e , we have to test the hypothesis:

$H_0 : \theta = \hat{\theta}$

If sampling distribution of θ is normal then,

$$z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \sim N(0,1)$$

Standard Normal Distribution compared with distribution when n = 5 and $n = 15$

The t-distribution has different shapes depending on the size of the sample. When the sample is quite small, for example, *if n* equals five, the height of the t-distribution is shorter than the normal distribution and the tails are wider. As *n* nears 30, however, the t-distribution approaches the normal distribution in shape.

The t-table

The t-table given is the probability integral of t-distribution. It gives over a range of values of v at different levels of significance. By selecting a particular degrees of freedom and level of significance, we determine the tabular value of *t.* We establish a null hypothesis, and if our computed *t* is greater than the tabular *t,* we reject the null hypothesis; if our computed tis smaller than the tabular *t,* we accept the null hypothesis.

Applications of t-distribution

The following are some important applications of the t-distribution :

(1) Test of Hypothesis about the population mean.

(2) Test of Hypothesis about the difference between two means with independent samples.

(3) Test of hypothesis about the difference between two means with dependent samples.

(4) Test of hypothesis about coefficient of correlation.

(l) Test of Hypothesis about the Population Mean (σunknown and sample size is small)

When the population distribution is normal and standard deviation σis unknown then the "t" statistic is defined as:

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

follows the Student's t-distribution with *(n - l) d.f.* where:

$\bar{x}$ = sample mean, μ = hypothesized population mean and n= sample size

and s is the standard deviation of the sample calculated by the formula:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

The null hypothesis to be tested is whether there is a significant difference between $\bar{x}$ and μ.

If the calculated value of *t* exceeds the table value of *t* at a specified level of significance, the null hypothesis is rejected and the difference between $\bar{x}$ and μ is regarded significant. If the calculated value of tis less than the table value, the difference between $\bar{x}$ and μ is not considered to be significant. It may be noted that this test is based on

n - 1 degrees of freedom.

Confidence Interval for the Population Mean

When sampling is from a normally distributed population with unknown σ,the 100 (1 - α) per cent confidence interval for the population is given by:

$$\bar{x} - t_{\alpha/2}(n-1)s/\sqrt{n} < \mu < \bar{x} + t_{\alpha/2}(n-1)s/\sqrt{n}$$

Example 1

Ten oil tins are taken at random from an automatic filling machine. The mean weight of the tins is 15.8 kg and standard deviation is 0.50 kg.Does the sample mean differ significantly from the intended weight of 16 kg?

Solution

Let the null hypothesis be that the sample mean weight is not different from the intended weight.

Given that $n = 10,\ \bar{x} = 15.8,\ s = 0.50,\ \mu = 16$

Using the t-test, we have

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{15.8 - 16}{0.5/\sqrt{10}} = \frac{-0.2}{0.158} = -1.266$$

The table value of t for 9 d.f. at 5% level of significance is 2.26. The computed value of t is smaller than the table value of t. Therefore, the difference is insignificant and the null hypothesis is accepted. Hence the difference between sample mean weight
and the intended weight is insignificant.

Example 2

Prices of shares (in $) of a company on the different days in a month were found to be:

66, 65, 69, 70,69, 71,70, 63, 64 and 68.

Test, whether the mean price of the shares in the month is 65.

Solution

We want to test the hypothesis:

$H_0 : \mu = 65$ *versus*
$H_0 : \mu \neq 65$

To test the above hypothesis, we compute;

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

From our calculator, $n = 10,\ \bar{x} = 67.5,\ s = 2.8,\ \mu = 65$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{67.5 - 65}{2.8/\sqrt{10}} = \frac{2.5}{0.89} = 2.81$$

The table value t for 9 degrees of freedom at 5% level of significance is 2.26. Since the computed value of t = 2.81 is greater than the table value, we reject the null hypothesis. Hence, the mean price of the shares in the month is not 65.

SPSS OUTPUT

```
T-TEST
  /TESTVAL=65
  /MISSING=ANALYSIS
  /VARIABLES=price
  /CRITERIA=CI(.95).
```

T-Test

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
price of shares	10	67.50	2.799	.885

One-Sample Test

	Test Value = 65					
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
price of shares	2.825	9	.020	2.500	.50	4.50

Conclusion

Since computed P-value = 0.02 < 0.05, we reject H_0 and conclude that the mean price of the shares in the month is not 65.

(2) Test of Hypothesis about the Difference between Two Means

In testing a hypothesis concerning the difference between the means of two normally distributed populations when the population variances are unknown, the *t-test* can be used in two types of cases;

(a) the case in which variances are equal, *i.e.*, $\sigma_1^2 = \sigma_2^2$, *(b)* the case in which variances are not equal, *i.e.*, $\sigma_1^2 \neq \sigma_2^2$.

(a) Case of equal variances

Let the null hypothesis be that there is no significant difference between the means of the two populations, *i.e.*, $H_0 : \mu_1 = \mu_2$. When the population variances (though unknown) are equal then the appropriate test statistic to be used is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

will follow t-distribution with $(n_1 + n_2 - 2)$ d.f., where $\bar{x}_1$ *and* $\bar{x}_2$ are sample means of sample 1 of size n_1 and sample 2 of size n_2 respectively; μ_1 and

μ_2 are the population means, and *s* is "pooled" estimate of the common population standard deviation obtained by pooling the data from both the samples as given below:

$$s=\sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}} \text{ where } s_1^2=\frac{\sum(x_1-\bar{x}_1)^2}{n_1-1}; \textit{ and } s_2^2=\frac{\sum(x_2-\bar{x}_2)^2}{n_2-1}$$

If the computed value of *t* is less than the table value of *t* at a specified level of significance,the null hypothesis is accepted and the difference between the two means is regarded asinsignificant. If the computed value of *t* is more than the table value of *t,* the null hypothesis is rejected and the difference between the sample means is regarded assignificant.

***b)* Case of unequal variances**

When the population variances are not equal*, i.e.* $\sigma_1^2 \neq \sigma_2^2$,we use the unbiased estimators s_1^2 and s_2^2 to replace σ_1^2 *and* σ_2^2 .In this case, the difficulty arisesbecausethe sampling distribution has large variability than the population variability.The statistic:

$$t=\frac{(\bar{x}_1-\bar{x}_2)-(\mu_1-\mu_2)}{s\sqrt{\frac{s_1^2}{n_1}+\frac{s_1^2}{n_2}}}$$

maynot strictly follow t-distribution but may be approximated byt-distribution with a modified value for the degrees of freedom given by

$$d.f.=\frac{\left[s_1^2/n_1+s_2^2/n_2\right]^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1-1}+\frac{\left(s_1^2/n_1\right)^2}{n_2-1}}$$

Example 3

Two different types of drugs A and B were tried on certain patients for increasing weight, 5 persons were given drug A and 7 persons were given drug B. The increase in weight (in pounds) is given below:

Drug A	8	12	13	9	3		
Drug B	10	8	12	15	6	8	11

Do the two drugs differ significantlywith regard to their effect in increasing weight?

Solution

Null hypothesis $H_0 : \mu_1 = \mu_2$, there is no significant difference in the efficacyof the two drugs. Thus

We want to test the hypothesis:

$H_0 : \mu_1 = \mu_2 \quad versus$

$H_0 : \mu_1 \neq \mu_2$

To test the above hypothesis, we compute;

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

From our calculator, $n_1 = 5,\ n_2 = 7,\ \bar{x}_1 = 9, \bar{x}_2 = 10, s_1 = 15.5,\ s_2 = 9$.

The pooled standard deviation is given by;

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(5-1)(15.5)^2 + (7-1)9^2}{5 + 7 - 2}} = \sqrt{11.6} = 3.406$$

Therefore;

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}} = \frac{9 - 10}{3.406}\sqrt{\frac{5 \times 7}{5 + 7}} = -\frac{1}{3.406} \times 1.708 = -0.5$$

$v = n_1 + n_2 - 2 = 5 + 7 - 2 = 10$. At $\alpha = 5\%$ level of significance,

$t_{\alpha/2}(10) = t_{0.025}(10) = 2.228$

The calculated value of *t* is lessthan the table value.Our null hypothesisis accepted.Hence, we conclude that there isno significantdifference in the efficacy of the twodrugsin the matter of increasingweight.

SPSS OUTPUT

```
T-TEST GROUPS=Drug(1 2)
  /MISSING=ANALYSIS
  /VARIABLES=weight
  /CRITERIA=CI(.95).
```

T-Test

Group Statistics

	Drug	N	Mean	Std. Deviation	Std. Error Mean
weigh	1	5	9.00	3.937	1.761
t	2	7	10.00	3.000	1.134

Independent Samples Test

		Levene's Test for Equality of Variances				
		F	Sig.	t	df	Sig. (2-tailed)
weight	Equal variances assumed	.192	.671	-.501	10	.6
	Equal variances not assumed			-.478	7.182	.6

Conclusion

Since computed P-value = 0.627 > 0.05, we accept H_0 and conclude that that there isno significantdifference in the efficacy of the twodrugsin the matter of increasingweight.

Example 4

Two salesmen A and Bare working in a certain district. From a sample survey conducted bythe Head Office,the following results were obtained. State whether there is any significant difference in the average sales between the two salesmen.

	A	B
No of sales	10	18
Average sales (in lakh Rs)	190	205
Standard deviation (in lakh Rs)	20	25

Solution

Null hypothesis; $H_0 : \mu_1 = \mu_2$ *i.e.,* there is no significant difference in the average sales between the two salesmen.

We want to test the hypothesis:

$H_0 : \mu_1 = \mu_2 \quad versus$

$H_0 : \mu_1 \neq \mu_2$

To test the above hypothesis, we compute;

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}} \text{ where}$$

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(9)(20)^2 + (17)25^2}{10 + 18 - 2}} = \sqrt{\frac{3600 + 10625}{26}} = 23.39$$

Therefore;

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}} = \frac{190 - 205}{23.39}\sqrt{\frac{10 \times 18}{10 + 18}} = \frac{15}{23.39} \times 2.54 = 1.63$$

$v = n_1 + n_2 - 2 = 10 + 18 - 2 = 26$. At $\alpha = 5\%$ level of significance,

$t_{\alpha/2}(26) = t_{0.025}(26) = 2.056$

The table value of t at 5% level ofsignificance for 26 *d.f.* is 2.056.The calculated value of t is less than the table value.The hypothesis holds true. Hence,we conclude that there isno significant difference in the average sales between the two salesmen.

Confidence Interval for the Difference between the Two Means

Two samples of sizes n_1 *and* n_2 are randomly and independently drawn from two normally distributed populations with unknowns but equal variances. The $100(1-\alpha)$ per cent confidence interval for $\mu_1 - \mu_2$ is given by:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}(n_1 + n_2 - 2)\, s / \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

(3) Test of hypothesis about the difference between two means with dependent samples (This is also known as paired t-test)

In the previous section, we assumed that the two random samples drawn from the two populations were independent. In many practical situations, this may not be true. The samples are *dependent* if they are paired so that each observation in one sample is associated with some particular observation in the second sample. Because of this property, the test we are going to use will be called *paired t-test.* In this test, it is necessary that the observations in the two samples be collected in the form called matched pairs. If two samples are dependent, they must have the same number of units. Instead of obtaining two random samples, we can get one random sample of pairs, and the two measurements associated with a pair will be related to each other. This kind of a problem arises in cases such as before and after type experiment or when observations are matched by rise or some other criterion. Suppose that two training methods are to be compared on the basis of average scores by management trainees divided into two equal size classes, one taught by each method. When experimental results are available, we test the null hypothesis that the means associated with the

twomethods are equal, i.e., $H_0 : \mu_1 = \mu_2$. The appropriate test statistic to be used here is

$$t = \frac{\bar{d}\sqrt{n}}{s}$$

follows *t-distribution with (n – 1) d.f.,* where $\bar{d}$ =mean of thedifferences is given by

$\bar{d} = \frac{\sum d}{n}$, s is the standard deviation of the differences and is given by:

$$s = \sqrt{\frac{\sum (d - \bar{d})^2}{n-1}} = \sqrt{\frac{\sum d^2 - n(\bar{d})^2}{n-1}}$$

and n is the number of paired observations in the samples.If the computed value of t (at a specified level of significance for a given number of degrees of freedom) is less than the table value of t, our null hypothesis is accepted,otherwise rejected.

Example 5

Ten persons were appointed in an officer cadre in an office. Their performance was noted by giving a test and the marks were recorded out of 100.Theywere given 4 monthstrainingand a test washeld and marks were recorded out of 100.

Employee	A	B	C	D	E	F	G	H	I	J
Before training	80	76	92	60	70	56	74	56	70	56
After training	84	70	96	80	70	52	84	72	72	50

By applying the t-test,can itbe concluded that the employees have benefited bythe training?

Solution

Let us take the null hypothesis that the employeeshave not benefited bythe training. Applying *t-test:*

Employee	Before (1st)	After (2nd)	d=(1st – 2nd)	d2
A	80	84	-4	16
B	76	70	6	36
C	92	96	-4	16
D	60	80	-20	400
E	70	70	0	0
F	56	52	4	16
G	74	84	-10	100
H	56	72	-16	256
I	70	72	-2	4
J	56	50	6	36
			$\sum d = -40$	$\sum d^2 = 880$

$t = \frac{\bar{d}\sqrt{n}}{s}$, where $\bar{d} = \frac{\sum d}{n} = \frac{-40}{10} = -4$,

$s = \sqrt{\frac{\sum d^2 - n(\bar{d})^2}{n-1}} = \sqrt{\frac{880-10(-4)^2}{9}} = \sqrt{\frac{880-160}{9}} = 8.944$. Therefore;

$t = \frac{\bar{d}\sqrt{n}}{s} = \frac{-4\sqrt{10}}{8.944} = -1.414$.

$v = n-1 = 10-1 = 9$. At $\alpha = 5\%$ level of significance, $t_{\alpha/2}(9) = t_{0.025}(9) = 2.262$.

The calculated value is less than the table value. Thenull hypothesis holdstrue.Hence,it can beconcluded that the employees have not benefited by the training.

SPSS OUTPUT

T-Test

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	before	69.00	10	12.083	3.821
	after	73.00	10	14.181	4.485

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	before & after	10	.779	.008

Paired Samples Test

	Paired Differences					t	df	S t
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			

Pair 1 before - after	-4.000	8.944	2.828	-10.398	2.398	-1.414	9	.

Conclusion from the SPSS Output

Since computed P-value = 0.191 > 0.05, the null hypothesis is accepted. Hence,it can beconcluded that the employees have not benefited by the training.

Example 6

Ten workers were given a training programme with a view to shorten their assembly time for a certain mechanism. The results of the time and motion studies before and after the training programme are given below:

Worker	1	2	3	4	5	6	7	8	9	10
First study (in mnts)	15	18	20	17	16	14	21	19	13	22
Second study (in mnts)	14	16	21	10	15	18	19	16	14	20

On the basis of this data, can it be concluded that the training programme has shortened the average assembly time?

Solution

Let ustakethenull hypothesisthat the training programme hasnot helped in reducing the average assembly time. Applying the t-test:

$$t = \frac{\bar{d}\sqrt{n}}{s}$$

Calculation for $\bar{d}$ *and* s

Employee	1st study	2nd study	d=(1st – 2nd)	d^2
1	15	14	1	1
2	18	16	2	4
3	20	21	-1	1
4	17	10	7	49
5	16	15	1	1
6	14	18	-4	16
7	21	19	2	4
8	19	16	3	9
9	13	14	-1	1
10	22	20	2	4
			$\sum d = 12$	$\sum d^2 = 90$

$t = \frac{\bar{d}\sqrt{n}}{s}$, where $\bar{d} = \frac{\sum d}{n} = \frac{12}{10} = 1.2$,

$s = \sqrt{\frac{\sum d^2 - n(\bar{d})^2}{n-1}} = \sqrt{\frac{90 - 10(1.2)^2}{9}} = \sqrt{10 - 1.6} = \sqrt{8.4} = 2.898$. Therefore;

$t = \frac{\bar{d}\sqrt{n}}{s} = \frac{1.2\sqrt{10}}{2.898} = 1.309$.

For $v = n - 1 = 10 - 1 = 9$. At $\alpha = 5\%$ level of significance, $t_{\alpha/2}(9) = t_{0.025}(9) = 2.262$

Since the computed value of *t* is less than the table value,the null hypothesis is accepted. Hence,the training programme has not shortened the average assembly time.

SPSS OUTPUT

T-Test

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	First	17.50	10	3.028	.957
	Second	16.30	10	3.302	1.044

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	First & Second	10	.584	.077

Paired Samples Test

		Paired Differences					t
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		
					Lower	Upper	
Pair 1	First - Second	1.200	2.898	.917	-.873	3.273	1.3

Conclusion from the SPSS Output

Since computed P-value = 0.223 > 0.05, the null hypothesis is accepted. Hence,the training programme has not shortened the average assembly time.

Confidence Interval for the Mean of the Difference

When the population for the mean of differences is normally distributed with unknown variance for dependent samples, a $100(1-\alpha)$ per cent confidence interval is given by:

$$\bar{d} \pm t_{\alpha/2}(n-1)\, s/\sqrt{n}$$

(4) Test of Hypothesis about Coefficient of Correlation

Case 1: Testing the hypothesis when the population coefficient of correlation equals zero, *i.e.,* $H_0 : \rho = 0$

Here the null hypothesisis that there isno correlation in the population, *i.e.,* $H_0 : \rho = 0$. The population coefficient of correlation ρ measuresthe degree of relationship between the variables.When $\rho = 0$, there is no statistical relationship between the variables.In order to test this hypothesis, it is necessary to know the sample coefficient of correlation *r (*which is the best estimate of ρ). The appropriate test statistic to be used here is given by:

$$t = \frac{r}{\sqrt{1-r^2}} \times \sqrt{n-2}$$

which followst-distribution with *n* - 2 degrees of freedom.

If the computed value of *t* is -greater than the table value of *t,* the null hypothesis is rejected which indicates that sample data provides sufficient evidence to indicate that $\rho \neq 0$. Hence,it can be concluded that there is alinear relationship between the variables.

Example 7

In a study of the relationship between expenditure (X) and annual salesvolume *(Y,*a sample of 10 firms yieldedthe coefficient ofcorrelation r = 0.93. Can we conclude on the basis of this data that X and Y are linearly related?

Solution

Thenull hypothesis is $H_0 : \rho = 0$, i.e.,there is no relationship between two variables. Using the t-test;

$$t = \frac{r}{\sqrt{1-r^2}} \times \sqrt{n-2} = \frac{0.93}{\sqrt{1-(0.93)^2}} \times \sqrt{10-2} = \frac{0.93}{\sqrt{0.14}} \times \sqrt{8} = \frac{0.93 \times 2.828}{0.374} = 7.03$$

The degrees of freedom or v = n – 2 = 10 – 2 = 8.

The table value of *t at* 5%level of significance for 8 *d.f. is 2.306*.Since the computed value ismuch greater value, thenull hypothesis is rejected.Hence,it maybe concluded that X and Y are linearly related.

Case 2: Testing the hypothesis when the population coefficient of correlation of other value than zero, *i.e.,* $H_0 : \rho = \rho_0$

In this case when $\rho \neq 0$, the test based on t-distribution will not be appropriate. In testing the hypothesis, the use of Fisher's z-transformation will be applicable.Here,*r* is transformed into zby

$$z = \frac{1}{2}\log_e\left[\frac{1+r}{1-r}\right]$$

Here, $\log_e$ is a *natural* logarithm. Common logarithms may be shifted to natural logarithms by multiplying by the factor 2.3026, i.e.,

$\log_e X = 2.3026 \log_{10} X$, where X isa positive integer.

Since $\frac{1}{2}(2.3026) = 1.1513$, the transformation formula may be used as :

$$z = \frac{1}{2}\log_e\left[\frac{1+r}{1-r}\right] = 1.1513\log_{10}\left[\frac{1+r}{1-r}\right]$$

Now, it can be shown that Z is normally distributed with mean

$z_\rho = \frac{1}{2}\log_e\left[\frac{1+\rho}{1-\rho}\right]$ and standard deviation , $\sigma_z = \frac{1}{\sqrt{n-3}}$

Therefore, to test the null hypothesis that $\rho = \rho_0$,the test statistic would be:

$$z = \frac{z - z_\rho}{\sigma_z}$$

which follows approximately the standard normal distribution. This test is more appropriate if sample size is large. The approximation is reasonably good if the sample size is at least 10.

Case 3 : Testing the hypothesis for the difference between two independent correlation coefficients

To test the hypothesis of two correlation coefficients derived from two separate samples, we have to compare the difference of the two corresponding values of zwith the standard error of that difference. In this case, the formula used will be:

$$z = \frac{z_1 - z_2}{\sigma_{z_1 - z_2}} = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}}}$$ where:

$$z_1 = \frac{1}{2}\log_e\left[\frac{1+r_1}{1-r_1}\right] = 1.1513\log_{10}\left[\frac{1+r_1}{1-r_1}\right] \text{ and } z_2 = \frac{1}{2}\log_e\left[\frac{1+r_2}{1-r_2}\right] = 1.1513\log_{10}\left[\frac{1+r_2}{1-r_2}\right].$$

If the absolute value of this statistic is greater than 1.96, the difference will be significant at 5 % level.

Example 8

The following data give sample size:

Sample size	Value of r
5	0.87
12	0.56

Test the significance of the difference between two values using Fisher's z-transformation.

Solution

Let the null hypothesis be that the experiment provides no evidence that the samples aredrawn from the same population.Applying z-test:

$$z = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}}}$$ where:

$$z_1 = 1.1513 \log_{10}\left[\frac{1+r_1}{1-r_1}\right] = 1.1513 \log_{10}\left[\frac{1+0.87}{1-0.87}\right] = 1.1513 \log_{10}\left[\frac{1.87}{0.13}\right] = 1.1513 \times 1.579 = 1.82$$

$$z_2 = 1.1513 \log_{10}\left[\frac{1+r_2}{1-r_2}\right] = 1.1513 \log_{10}\left[\frac{1+0.56}{1-0.56}\right] = 1.1513 \log_{10}\left[\frac{1.56}{0.44}\right] = 1.1513 \times 0.5495 = 0.63$$

$$z = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}}} = \frac{1.82 - 0.63}{\sqrt{\frac{1}{5-3} + \frac{1}{12-3}}} = \frac{1.19}{\sqrt{0.5 + 0.1}} = \frac{1.19}{\sqrt{0.6}} = \frac{1.19}{0.77} = 1.54$$

Since the computed value of zis lessthan the table value of z at 5% level of significance = 1.96, therefore,the null hypothesis is accepted.Hence,the experiment provides no evidence against the hypothesis that the samples are drawn from the same population.

Example 9

Suppose we want to test whether *r* = 0.884 for a pair of 20 observations is significantly different from a hypothesized value $\rho = 0.92$.

Solution

We transform *r* into zby

$$z = 1.1513 \log_{10}\left[\frac{1+0.884}{1-0.884}\right] = 1.3938$$

The distribution of zis approximately normal around the hypothesized value

$\rho_0 = 0.92 = z_{\rho_0}$, where

$$z_{\rho_0} = 1.1513 \log_{10}\left[\frac{1+0.92}{1-0.92}\right] = 1.5890$$

The distribution of z hasa standard deviation

$$\sigma_z = \sqrt{\frac{1}{n-3}} = \sqrt{\frac{1}{20-3}} = \sqrt{\frac{1}{17}} = 0.2425$$

Therefore, the statistic is

$$z = \frac{1.3938 - 1.5890}{0.2425} = -0.80$$

From the table of areas for the normal curve, we find that at = 5% level of significance, z = 1.96. Since computed $|z| = 0.8 < 1.96$, we accept H_0 and conclude that $\rho = 0.92$.

The F - Distribution

The *F-distribution* is named in honour of R.A. Fisher who *first studied it* in 1922 is usually defined in terms of the ratio of the variances of two normally distributed quantity

$$\frac{s_1^2 / \sigma_1^2}{s_2^2 / \sigma_2^2}$$

is distributed as *F-distribution* with $v_1 = n_1 - 1$ *and* $v_2 = n_2 - 1$ degrees of freedom, where

$s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1}$ is the unbiased estimator of σ_1^2 and $s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1}$ is the unbiased estimator of σ_2^2.

If $\sigma_1^2 = \sigma_2^2$,then the statistic;

$$F = \frac{s_1^2}{s_2^2}$$

follows *F-distribution* with $v_1 = n_1 - 1$ *and* $v_2 = n_2 - 1$ degrees of freedom.

The F-distribution sometimes is also called *VarianceRatio*distribution.

The F-distribution depends on the degrees of freedom v_1 for the numerator and v_2 for the denominator.Therefore, the parameters for F-

distribution are v_1 *and* v_2.For different values of v_1 *and* v_2, we shall have different distributions.

The probability density function of *F-distribution* is given by:

$$f(F) = Y_0 \frac{F^{\frac{v_1}{2}-1}}{\left[1+\frac{v_1}{v_2}\right]^{(v_1+v_2)/2}}, \quad 0 \le F \le \infty$$ where Y_0 is a constant depending on values v_1 *and* v_2 such that the area under the curve is unity. A typical F-distribution is given below:

Some of the important properties of the F-distribution are given below:

1) The F-distribution is positively skewed and its skewness decreases with increase in v_1 *and* v_2.

(2) The value of *F* must always be positive or zero since variances are squares and can never assume negative values. Its value will always lie between 0 and ∞.

(3) The shape of the F-distribution depends upon the number of degrees of freedom.

(4) The areas in the left-hand side of the distribution can be found by taking the reciprocal of F values corresponding to the right-hand

side,when the number of degrees of freedom in the numerator and in the denominator are interchanged. This is also known as *reciprocal property and can be expressed as:*

$$F_{1-\alpha}(v_1, v_2) = \frac{1}{F_{\alpha}(v_2, v_1)}$$

where the symbols have their usual meanings. This property is of great help when we want to know the lower tail *F* values from corresponding upper tail *F* values which are given in the appendix.

Testing of Hypothesis for Equality of two Variances

The test of equality of two population variances is based on the variances in two independently selected random samples drawn from two normal populations. Under the null hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$;

$F = \frac{s_1^2 / \sigma_1^2}{s_2^2 / \sigma_2^2}$ reduces to $F = \frac{s_1^2}{s_2^2}$.

which follows F-distribution with v_1 *and* v_2 degrees of freedom. It is convenient to place larger sample variance in the numerator for computational purpose. If we do so, the ratio of the sample variance will be equal to or greater than one.

If the computed value of *F* exceeds the table value of F,we reject the null hypothesis, *i.e.,* the alternative hypothesis is accepted.

Example 10

Two sources of rawmaterials are under consideration bya company. Both sources seem to have similar characteristics but the company isnot sure about their respective uniformity. A sample of 10 lots from source *A* yields a variance

of 225 and asample of 11 lots from source Byields avariance of 200.Is it likely that the variance of source Ais significantly greater than the variance of source Bat $\alpha = 0.01$?

Solution

Null hypothesisis $H_0 : \sigma_1^2 = \sigma_2^2$, i.e., the variances of source A and that of source Bare the same. The *F* statistic to be used here is:

$$F = \frac{s_1^2}{s_2^2} = \frac{225}{200} = 1.1$$

The table value of F for $v_1 = 9$ *and* $v_2 = 10$ at 1% level of significance is 4.94. Since the computed value of *F* issmaller than the table value of F,the null hypothesis is accepted. Hence,the population variances of the two populations are same.

Confidence Interval for the Ratio of Two Variances

A $100(1-\alpha)$ per cent confidence interval for the ratio of the variances of two normally distributed populations is given by:

$$\frac{s_1^2 / s_2^2}{F_{1-\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2 / s_2^2}{F_{\alpha/2}(v_1, v_2)}$$

where the symbols have their usual meaning.

CHAPTER SIX: CHI-SQUARE TESTS

In the previous chapters, it was necessary to make certain assumptions about the population from which the samples were drawn. In many of

the statistical tests, we had to assume that the samples came from normal populations. When this assumption cannot be justified, it is necessary to use procedures that do not require that these conditions be met. These procedures are generally referred to as non-parametric methods. In this chapter, we will discuss the χ^2 – test which belongs to this category. The χ^2 (pronounced as Chi-square) test is based on χ^2 – distribution which was first introduced by Karl Pearson in the year 1900.

The Chi-squareDistribution

For large sample size, the sampling (probability) distribution of χ^2 can be closely approximated by a continuous curve known as the Chi-square distribution. The probability function of χ^2 – distribution is given by:

$f(\chi^2) = c(\chi^2)^{\frac{v}{2}-1} e^{-\frac{\chi^2}{2}}$ where:

e = 2.71828, v = number of degrees of freedom and c = a constant depending only on v. The Chi-square distribution has only one parameter v, the number of degrees of freedom.

Important Properties of Chi-square Distribution

(1) χ^2 – distribution is a continuous probability distribution which has the value zero at its lower limit and extends to infinity in the positive direction. Negative values of χ^2 is not possible.

(2) The exact shape of the χ^2 – distribution depends upon the number of degrees of freedom v. For different values of v, we shall have different shapes of the distribution. In general, when v is small, the shape of the curve is skewed to the right and as v gets larger, the

distribution becomes more and more symmetrical and can be approximated by the normal distribution.

(3) The mean of the χ^2 – distribution is given by the degrees of freedom, i.e., $E(\chi^2) = v$ and variance is twice the degrees of freedom, i.e., $Var(\chi^2) = 2v$.

(4) As v gets larger, χ^2 – distribution approaches the normal distribution with mean = v and variance = $2v$.

(5) The sum of independent χ^2 variate is also a χ^2 variate. Therefore, if χ_1^2 is a variate with v_1 degrees of freedom and χ_2^2 is a χ^2 variate with v_2 degrees of freedom independent of χ_1^2, then their sum $\chi_1^2 + \chi_2^2$ is also a χ^2 variate with $v_1 + v_2$ degrees of freedom. This property is known as the additive property of χ^2.

Chi-square test

The χ^2 – test is one of the simplest and most widely used non-parametric tests in statistical work. It makes no assumptions about the population being sampled. The quantity χ^2 describes the magnitude of discrepancy between theory and observation, i.e., with the help of χ^2 – test we can know whether a given discrepancy between theory and observation can be attributed to chance or whether it results from the inadequacy of the theory to fit the observed facts. The formula for computing the Chi-square test is:

$\chi^2 = \sum \frac{(O-E)^2}{E}$, where:

O = observed frequency and E = expected or theoretical frequency.

The calculated value of χ^2 is compared with the table value of χ^2 for given degrees of freedom at specified level of significance. If the calculated value of χ^2 is greater than the table value, the difference between theory and observation is considered to be significant. On the other hand, if the calculated value of χ^2 is less than the table value, the difference between theory and observation is not considered to be significant.

The number of degrees of freedom is described as the number of observations that are free to vary after certain restrictions have been imposed on the data.

For a uniform distribution, we place one restriction on the expected distribution – the total of sample observations.

In a contingency table, the numbers of degrees of freedom are calculated as:

$(r-1)\times(c-1)$, where r = number of rows in the table and c = number of columns in the table.

Conditions for the Application of χ^2 – Test

The following five basic conditions must be met in order for Chi-square analysis be applied:

(1) The experimental data (sample observation) must be independent of each other.

(2) The sample data must be drawn at random from the target population.

(3) The data should be expressed in original units for convenience of comparison and not in percentage or ratio form.

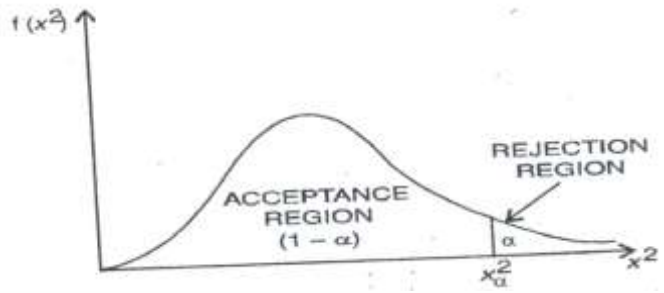
(4) The sample should contain at least 50 observations.

(5) There should not be less than five observations in any one cell (each data entry is known as a cell).

Use of Chi-square Table

To facilitate its many applications, the Chi-square distribution has been extensively

tabulated. The table of areas found in Chi-square tables gives values of χ^2 for variousprobabilities and various degrees of freedom. The value of is given in the columnheadings, the degrees of freedom,v are given in the rows and the body of the tablegives the χ^2 values. As depicted in the following figure, the value of χ^2 in the Chi-square table are given for various combinations of v and $1-\alpha$.



Grouping when Frequencies are small

If small theoretical frequencies occur (less than 10 and certainly not less than 5), it is generally possible to overcome the difficulty by grouping two or more classes together. In other words, one or more classes with theoretical frequencies less than 5 may be combined into a single category before calculating the difference between observed and expected frequencies. The number of degrees of freedom would be

determined with the number of observations after the regrouping. This would be clear from the following:

Observed frequencies	364	376	218	89	33	13	2	1
Expected frequencies	339	397	234	92	27	6	1	0

The last three classes should be combined together. After grouping, the position would be as follows:

Observed frequencies	364	376	218	89	33	16
Expected frequencies	339	397	234	92	27	7

The degrees of freedom would now be 6 – 1 = 5.

Some important applications of χ^2 – test are discussed in detail below:

(1) Tests of Hypothesis Concerning Variance

In testing hypothesis about the variance of a normally distributed population, the null hypothesis is $H_0 : \sigma^2 = \sigma_0^2$ against the alternative hypothesis $H_1 : \sigma^2 \neq \sigma_0^2$ where σ_0^2 is some specified value of the population variance.

To test the above hypothesis we compute $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ where s^2 is computed from a random sample of size n.

Conclusion

If computed $\chi^2 > \chi^2_{\alpha/2}(n-1)$, we reject H_0, otherwise we accept H_0.

Example 1

Weights in kilograms of 10 shipments are given below:

38, 40, 45, 53, 47, 43, 55, 48, 52, 49.

Can we say that the variance of the distribution of weight of all shipments from which the above sample of 10 shipments was drawn is equal to 20 square kilogram?

Test at □ □ = 5% level of significance.

Solution

We want to test the hypothesis:

$H_0 : \sigma^2 = 20$ against the alternative hypothesis $H_1 : \sigma^2 \neq 20$.

To test the above hypothesis, we compute:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}.$$

From our calculator, $s^2 = 31.111$ and $n = 10$. Hence

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{9 \times 31.1111}{20} = \frac{280}{20} = 14$$

The table value of χ^2 for 9 d.f. at 5% level of significance is 16.919.

Conclusion

Since computed $\chi^2 = 14 < \chi^2_{\alpha/2}(n-1) = 16.919$, we accept H_0. Hence, we conclude that the data are consistent with the hypothesis that the variance of the distribution of weights of all shipments in the population is 20 kilograms.

(2) Test of Independence

One of the most frequent uses of χ^2 is for testing the null hypothesis that two criteria of classification are independent. They are independent if the distribution of one criterion in no way depends on the distribution of the other criterion. If they are not independent, there is an association between the two criteria. In the test of independence, the population and sample are classified according to some attributes. The test will indicate only, whether or not any dependency relationship exists between the attributes. It will not indicate the degree of association or the direction of dependency.

To conduct the test, a sample is drawn from the sample of interest and observed frequencies are cross-classified according to the two criteria. The cross-classification can be conveniently displayed by means of a table called a *contingency table*. Let us designate the two attributes as A and B where, attribute A is assumed to have *r* categories and attribute B is assumed to have *c* categories. Furthermore, assume the total number of observations in the problem is N. A representation of these observations is shown below in a table where O_{ij} represents the observation in the ith row and j^{th} column. Such a table in the matrix form is called a contingency table and is shown below.

	Attribute B							
Attribute A	B_1	B_2	B_3	…	B_j	…	B_c	Total
A_1	O_{11}	O_{12}	O_{13}	…	O_{1j}	…	O_{1c}	R_1
A_2	O_{21}	O_{22}	O_{23}	…	O_{2j}	…	O_{2c}	R_2
A_3	O_{31}	O_{32}	O_{33}	…	O_{3j}	…	O_{3c}	R_3
⋮	⋮	⋮	⋮		⋮		⋮	⋮
A_i	O_{i1}	O_{i2}	O_{i3}	…	O_{ij}	…	O_{ic}	R_i
⋮	⋮	⋮	⋮		⋮		⋮	⋮
A_r	O_{r1}	O_{r2}	O_{r3}	…	O_{rj}	…	O_{rc}	R_r
Total	C_1	C_2	C_3	…	C_j	…	C_c	N

In the table, R_i is the total of ith row and C_j is the total of jth column. The frequencies in these cells are termed as *cell frequencies* and the totals of the frequencies in each of the rows (R_i) and columns (C_j) are termed as *marginal frequencies*.

Expected cell frequencies are computed according to the multiplicative rule of probability. If two events are independent, the probability of

their joint occurrence is equal to the product of their individual probabilities. Applying this rule to a contingency table, it is equivalent to say that, if two criteria of classification are independent, a joint probability is equal to the product of the two corresponding marginal probabilities. Thus, the expected cell frequencies are given by the formula:

$$E_{ij} = \frac{R_i}{N} \times \frac{C_j}{N} \times N = \frac{R_i \times C_j}{N}$$

To conduct the test, same χ^2 is employed, i.e.,

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

will follow χ^2 - distribution with $v = (r-1) \times (c-1)$ degrees of freedom.

While applying the test, the null hypothesis is that the two attributes are independent.

If the calculated value of χ^2 is less than the table value at a specified level of significance, the null hypothesis holds true. If the calculated value of χ^2 is greater than the table value at a specified level of significance, the null hypothesis is rejected, i.e., the two attributes are associated.

Example 2

A sample of 200 persons with a particular disease was selected. Out of these, 100 were given a drug and the others were not given any drug. The results are as follows:

	Number of Persons		
	Drug	No Drug	Total
Cured	65	55	120
Not cured	35	45	80
Total	100	100	200

Tests, whether the drug is effective or not.

Solution

Let us take the null hypothesis that the drug is not effective in curing the disease. Applying the χ^2- tests: the expected cell frequencies are computed as follows:

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{120 \times 100}{200} = 60; \qquad E_{12} = \frac{R_1 \times C_2}{N} = \frac{120 \times 100}{200} = 60$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{80 \times 100}{200} = 40; \qquad E_{22} = \frac{R_2 \times C_2}{N} = \frac{80 \times 100}{200} = 40$$

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(65-60)^2}{60} + \frac{(55-60)^2}{60} + \frac{(35-40)^2}{40} + \frac{(45-40)^2}{40}$$
$$= 0.417 + 0.417 + 0.625 + 0.625 = 2.084$$

$$v = (r-1) \times (c-1) = (2-1) \times (2-1) = 1 \times 1 = 1$$

At □ = 5% level of significance; $\chi^2_{0.05}(1) = 3.84$

Conclusion

Since computed $\chi^2 = 2.084 < \chi^2_{0.05}(1) = 3.84$, we accept H_0. Hence, we conclude that the drug is not effective in curing the disease.